

Module 05: Subnetting Mechanics and Classful Subnet Design

Unit 3: Network Layer

Subnetting & Bit Borrowing

Concept of Subnetting, Host-Bit Borrowing, Subnet Mask Math, Block Size Formula, Step-by-Step Numerical Calculations & Bitwise AND Subnet ID Extraction.

Subnetting Architecture: Bit Borrowing & Subnet ID Formation

1. Original Class C IP Structure (/24 Mask: 255.255.255.0)

Network ID (24 Bits) - Fixed by ISP / IANA

Host ID (8 Bits) = 256 IPs

Borrow s = 2 Bits from Host ID

2. Subnetted Structure (/26 Mask: 255.255.255.192)

Network ID (24 Bits)

Subnet (2 Bits)

Host ID (6 Bits)
2^6 - 2 = 62 Hosts

Bitwise AND: Router Subnet Extraction

IP: 192.168.1.75 11000000.10101000.00000001.01001011
Mask: 255.255.255.192 11111111.11111111.11111111.11000000
AND: 192.168.1.64 11000000.10101000.00000001.01000000
Result: Packet belongs to Subnet 1 (Range: .64 to .127)

4 Subnets Generated (Jump Size = 64)

Subnet ID	First Usable IP	Last Usable IP	Broadcast ID
.0 (Sub 0)	192.168.1.1	192.168.1.62	192.168.1.63
.64 (Sub 1)	192.168.1.65	192.168.1.126	192.168.1.127
.128 (Sub 2)	192.168.1.129	192.168.1.190	192.168.1.191
.192 (Sub 3)	192.168.1.193	192.168.1.254	192.168.1.255

1. Fundamental Formulas

Subnet Engineering Rules

Number of Subnets = 2^s (where s = borrowed bits)

Valid Hosts per Subnet = 2^h - 2 (where h = remaining host bits)

Block Size (Jump Size) = 256 - Mask Octet Value = 2^h

Subnet ID = Destination IP Bitwise AND Subnet Mask

2. Subnet Allocation Table (Class C /26 Example: 193.1.2.0)

Subnet	Subnet ID	First Usable IP	Last Usable IP	Directed Broadcast
Subnet 0	193.1.2.0	193.1.2.1	193.1.2.62	193.1.2.63
Subnet 1	193.1.2.64	193.1.2.65	193.1.2.126	193.1.2.127
Subnet 2	193.1.2.128	193.1.2.129	193.1.2.190	193.1.2.191
Subnet 3	193.1.2.192	193.1.2.193	193.1.2.254	193.1.2.255